

TED (15) – 5212
REVISION – 2015

Reg. No.....
Signature.....

MODELQUESTION PAPER

PROCESS CONTROL

(Maximum Marks: 100)

(Time: 3 hours)

PART – A

(Maximum Marks: 10)

Marks

I. (Answer the following questions in one or two sentences. Each question carries 2 marks)

1. Define the term Process
2. State Neutral Zone
3. Interpret the term Integral Time
4. Define Actuator
5. State the term Data Transmission (5×2 =10)

PART – B

(Maximum Marks: 30)

II. (Answer *any five* of the following questions. Each question carries 6 marks)

1. Explain Controlled Variable, Manipulated Variable and Load Variable
2. Describe Multi Position Control Mode
3. Explain Pneumatic Error Detector
4. Draw the structure of Butterfly Valve
5. Explain the working of Limit Switch
6. Explain Position Telemetry system
7. Describe the advantages of Field Bus (5×6 = 30)

PART – C

(Maximum Marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

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| III | 1. Explain the Elements of Process Control Loop | 8 |
| | 2. Describe Flow Process Control system with example | 7 |

OR

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| IV | 1. Explain Human Aided Control | 8 |
| | 2. Describe Process Characteristics | 7 |

UNIT – II

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| V | 1. With diagram describe ON – OFF Control Mode | 8 |
| | 2. Briefly describe Electronic PI Control mode | 7 |

OR

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| VI | 1. Briefly explain Pneumatic Proportional Controller | 8 |
| | 2. Explain Integral Control Mode | 7 |

UNIT – III

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| VII | 1. Explain the working of Pneumatic Actuator | 8 |
| | 2. Explain the working of Air Pressure Regulator | 7 |

OR

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| VIII | 1. Explain the operation of I/P Converter | 8 |
| | 2. Explain the Block diagram of Final Control Operation | 7 |

UNIT – IV

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| IX | 1. Explain the operation of Force Balance Current Telemetry system | 8 |
| | 2. Explain the Profibus Communication Standard | 7 |

OR

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| X | 1. Describe Functional elements of Foundation Field Bus | 8 |
| | 2. Explain the working of HART Communication | 7 |
